
Math 31

Advanced Algebra

Math 31 Course at a Glance

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

Review Content

REVIEW	SECTIONS	No mapped textbook section
M31-REV-001	I can calculate a percentage, percent increase, or percent decrease.	
M31-REV-002	I can simplify an expression using exponent properties.	
M31-REV-003	I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.	
M31-REV-004	I can expand a product of two binomials.	

Linear Functions and Function Notation

REQUIRED	SECTIONS	3.1, 3.2, 3.3, 4.1, 4.2
M31-LIN-001	I can interpret function notation as a statement about input and output.	
M31-LIN-002	I can determine whether a graph or table represents a function.	
M31-LIN-003	I can evaluate a function or solve for an input using a graph, table, or formula.	
M31-LIN-004	I can identify independent and dependent variables in a relationship.	
M31-LIN-005	I can calculate average rate of change from two points.	
M31-LIN-006	I can interpret interval notation and identify increasing or decreasing intervals.	
M31-LIN-007	I can determine total change from an average rate of change over an interval.	
M31-LIN-008	I can determine and interpret the units of an average rate of change.	
M31-LIN-009	I can identify a linear function from a table, graph, or formula.	
M31-LIN-010	I can interpret the slope and intercept of a linear model.	
M31-LIN-011	I can determine the intersection of two linear relationships graphically or algebraically.	
M31-LIN-012	I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.	

Function Behavior and Structure

REQUIRED	SECTIONS	3.1, 3.2, 3.3, 3.4, 3.5, 3.7
M31-FUN-001	I can determine whether a relation is a function.	
M31-FUN-002	I can evaluate a function from a formula, graph, or table.	
M31-FUN-003	I can determine domain and range from a formula, graph, or table.	
M31-FUN-004	I can determine algebraic domain restrictions caused by denominators and even roots.	
M31-FUN-005	I can graph a piecewise function from its formula.	
M31-FUN-006	I can write a piecewise formula that represents a graph.	
M31-FUN-007	I can graph a transformed function using shifts, stretches, compressions, and reflections.	
M31-FUN-008	I can describe how a transformation affects domain, range, or asymptotes.	
M31-FUN-009	I can evaluate a composition from formulas, graphs, or tables.	
M31-FUN-010	I can write a formula for a composition of two functions.	
M31-FUN-011	I can find an inverse function algebraically.	
M31-FUN-012	I can evaluate and interpret an inverse from a graph, formula, or table.	
M31-FUN-013	I can determine and interpret the units of an inverse function.	
M31-FUN-014	I can explain how the domain and range of a function relate to those of its inverse.	
M31-FUN-015	I can determine the range of a function by analyzing the domain of its inverse.	
M31-FUN-016	I can verify algebraically that two functions are inverses.	
M31-FUN-017	I can interpret the meaning and units of a composition.	
M31-FUN-018	I can classify a graph or table as concave up or concave down using changes in average rate of change.	
M31-FUN-019	I can determine whether a function is invertible using the horizontal line test.	
M31-FUN-020	I can graph an inverse function as a reflection across the line $y = x$.	

Quadratic Functions and Equations

REQUIRED	SECTIONS	2.5, 5.1
M31-QUAD-001	I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.	
M31-QUAD-002	I can connect standard, factored, and vertex forms to features of a quadratic graph.	
M31-QUAD-003	I can rewrite a quadratic function in a form that reveals a requested feature.	
M31-QUAD-004	I can solve a quadratic equation by factoring or using the quadratic formula.	
M31-QUAD-005	I can graph a quadratic function using its algebraic structure and key features.	
M31-QUAD-006	I can construct a quadratic function from specified zeros, a vertex, or graphical information.	
M31-QUAD-007	I can create and interpret a quadratic trajectory model.	

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Exponential Functions and Models

REQUIRED	SECTIONS 6.1, 6.2, 6.7
M31-EXP-001	I can distinguish linear change from exponential change using differences, ratios, or percent change.
M31-EXP-002	I can identify initial value and growth or decay factor in an exponential model.
M31-EXP-003	I can write an exponential function from a table, graph, or context.
M31-EXP-004	I can evaluate and interpret an exponential model.
M31-EXP-005	I can calculate growth under periodic compounding.
M31-EXP-006	I can calculate growth under continuous compounding.
M31-EXP-007	I can determine an exponential formula from two points or other sufficient data.
M31-EXP-008	I can determine domain, range, intercepts, and asymptotes of an exponential function.
M31-EXP-009	I can compare the growth rates of two exponential functions.
M31-EXP-010	I can determine the percent growth or decay represented by a continuous exponential model.

Logarithmic Functions and Models

REQUIRED	SECTIONS 6.3, 6.4, 6.5, 6.6, 6.7
M31-LOG-001	I can convert between logarithmic and exponential forms.
M31-LOG-002	I can evaluate a logarithm using its exponential meaning.
M31-LOG-003	I can expand or condense an expression using product, quotient, and power properties.
M31-LOG-004	I can solve an exponential equation using logarithms.
M31-LOG-005	I can solve a logarithmic equation using exponential form and check domain restrictions.
M31-LOG-006	I can calculate and interpret doubling time or half-life.
M31-LOG-007	I can determine domain, range, intercepts, and asymptotes of a logarithmic function.
M31-LOG-008	I can explain how exponential and logarithmic functions undo one another.
M31-LOG-009	I can solve a multi-step exponential or logarithmic equation.
M31-LOG-010	I can determine an intersection of exponential functions graphically or algebraically.
M31-LOG-011	I can rewrite a discrete exponential model as an equivalent continuous exponential model.
M31-LOG-012	I can solve and interpret an exponential or logarithmic modeling problem.

Extension Content

EXTENSION	SECTIONS 3.6
M31-EXT-001	I can analyze an absolute-value function using transformations.
M31-EXT-002	I can determine even or odd symmetry from a graph or formula.
M31-EXT-003	I can interpret a contextual logarithmic scale.

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Review Content

REVIEW

UNIT TEXTBOOK COVERAGE

No mapped textbook section

M31-REV-001

I can calculate a percentage, percent increase, or percent decrease.
TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

R	Calculate a percentage of a quantity or determine what percent one quantity is of another. (inherited from middle school) SK-NUM-PERCENT-CALCULATE · Reinforce
R	Calculate and interpret percent increase or decrease. (inherited from middle school) SK-NUM-PERCENT-CHANGE · Reinforce

M31-REV-002

I can simplify an expression using exponent properties.
TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

R	Rewrite a negative exponent using a reciprocal. (first introduced in Math 21) SK-EXP-NEGATIVE · Reinforce
R	Apply the power rule for exponents. (first introduced in Math 12) SK-EXP-POWER · Reinforce
R	Apply the product rule for exponents. (first introduced in Math 12) SK-EXP-PRODUCT · Reinforce
R	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Reinforce
R	Apply the zero-exponent rule. (first introduced in Math 21) SK-EXP-ZERO · Reinforce

M31-REV-003

I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.
TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

R	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Reinforce
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M31-REV-004

I can expand a product of two binomials.
TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Apply
R	Multiply two binomials using distribution. (first introduced in Math 12) SK-POLY-BINOMIAL-MULTIPLY · Reinforce

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Linear Functions and Function Notation

REQUIRED

UNIT TEXTBOOK COVERAGE

3.1 Functions and Function Notation; 3.2 Domain and Range; 3.3 Rates of Change and Behavior of Graphs; 4.1 Linear Functions; 4.2 Modeling with Linear Functions

M31-LIN-001	I can interpret function notation as a statement about input and output. TEXTBOOK SECTIONS 3.1 Functions and Function Notation
SUPPORTING SKILLS	
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Identify the input and output represented by function notation. (first introduced in Math 12) SK-FUNC-NOTATION-READ · Apply
I	Use function notation to express and solve an input-output question. SK-FUNC-NOTATION-SOLVE · Introduce

M31-LIN-002	I can determine whether a graph or table represents a function. TEXTBOOK SECTIONS 3.1 Functions and Function Notation
SUPPORTING SKILLS	
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
D	Apply the vertical line test to a graph. (first introduced in Math 12) SK-FUNC-VERTICAL-LINE · Deepen

M31-LIN-003	I can evaluate a function or solve for an input using a graph, table, or formula. TEXTBOOK SECTIONS 3.1 Functions and Function Notation
SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Use function notation to express and solve an input-output question. (first introduced in Math 31) SK-FUNC-NOTATION-SOLVE · Apply
D	Solve for an input associated with a specified output. (first introduced in Math 21) SK-FUNC-SOLVE-INPUT · Deepen

M31-LIN-004	I can identify independent and dependent variables in a relationship. TEXTBOOK SECTIONS 3.1 Functions and Function Notation
SUPPORTING SKILLS	
I	Identify independent and dependent variables. SK-FUNC-INDEPENDENT-DEPENDENT · Introduce
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply

M31-LIN-005	I can calculate average rate of change from two points. TEXTBOOK SECTIONS 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions
SUPPORTING SKILLS	
D	Calculate average rate of change over an interval. (first introduced in Math 21) SK-FUNC-AROC · Deepen
A	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Apply

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

M31-LIN-006

I can interpret interval notation and identify increasing or decreasing intervals.
TEXTBOOK SECTIONS 3.2 Domain and Range; 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

I	Describe increasing or decreasing behavior over intervals. SK-FUNC-INTERVAL-BEHAVIOR · Introduce
D	Translate between inequality and interval notation. (first introduced in Math 12) SK-NOT-INTERVAL · Deepen

M31-LIN-007

I can determine total change from an average rate of change over an interval.
TEXTBOOK SECTIONS 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

A	Calculate average rate of change over an interval. (first introduced in Math 21) SK-FUNC-AROC · Apply
I	Use average rate of change and interval length to calculate total change. SK-FUNC-AROC-TOTAL-CHANGE · Introduce
A	Add, subtract, multiply, and divide signed integers. (inherited from middle school) SK-NUM-INTEGGER-OPERATE · Apply

M31-LIN-008

I can determine and interpret the units of an average rate of change.
TEXTBOOK SECTIONS 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

I	Determine rate units from input- and output-axis units. SK-FUNC-AROC-UNITS · Introduce
D	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Deepen

M31-LIN-009

I can identify a linear function from a table, graph, or formula.
TEXTBOOK SECTIONS 4.1 Linear Functions; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

I	Use constant differences or ratios to distinguish linear and exponential patterns. SK-EXP-DIFFERENCE-RATIO · Introduce
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen
D	Use tabular patterns to predict graph shape and behavior. (first introduced in Math 21) SK-REP-TABLE-GRAPH · Deepen

M31-LIN-010

I can interpret the slope and intercept of a linear model.
TEXTBOOK SECTIONS 4.1 Linear Functions; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

A	Plot an intercept accurately on a coordinate plane. (first introduced in Math 12) SK-LIN-PLOT-INTERCEPT · Apply
A	Determine slope from a graph. (first introduced in Math 12) SK-LIN-SLOPE-GRAPH · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M31-LIN-011

I can determine the intersection of two linear relationships graphically or algebraically.
TEXTBOOK SECTIONS 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

A	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Apply
A	Add equations to eliminate a variable. (first introduced in Math 12) SK-SYS-ELIMINATE · Apply
A	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Apply
A	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Apply

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M31-LIN-012	I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point. TEXTBOOK SECTIONS4.1 Linear Functions
SUPPORTING SKILLS	
A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply
A	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Apply
D	Recognize that distinct parallel lines have equal slopes. (first introduced in Math 12) SK-LIN-PARALLEL-SLOPE · Deepen
D	Determine the negative reciprocal of a nonzero slope. (first introduced in Math 12) SK-LIN-PERPENDICULAR-SLOPE · Deepen

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Function Behavior and Structure

REQUIRED

UNIT TEXTBOOK COVERAGE

3.1 Functions and Function Notation; 3.2 Domain and Range; 3.3 Rates of Change and Behavior of Graphs; 3.4 Composition of Functions; 3.5 Transformation of Functions; 3.7 Inverse Functions

M31-FUN-001

I can determine whether a relation is a function.
TEXTBOOK SECTIONS 3.1 Functions and Function Notation

SUPPORTING SKILLS

D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Apply the vertical line test to a graph. (first introduced in Math 12) SK-FUNC-VERTICAL-LINE · Apply

M31-FUN-002

I can evaluate a function from a formula, graph, or table.
TEXTBOOK SECTIONS 3.1 Functions and Function Notation

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Solve for an input associated with a specified output. (first introduced in Math 21) SK-FUNC-SOLVE-INPUT · Apply

M31-FUN-003

I can determine domain and range from a formula, graph, or table.
TEXTBOOK SECTIONS 3.2 Domain and Range

SUPPORTING SKILLS

I	Determine domain restrictions from denominators, even roots, and logarithms. SK-FUNC-DOMAIN-FORMULA · Introduce
D	Read domain and range from a graph. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-GRAPH · Deepen
D	Read domain and range from a table. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-TABLE · Deepen

M31-FUN-004

I can determine algebraic domain restrictions caused by denominators and even roots.
TEXTBOOK SECTIONS 3.2 Domain and Range

SUPPORTING SKILLS

A	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Apply
A	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Apply
D	Require an even-root radicand to be nonnegative when determining domain. (first introduced in Math 21) SK-FUNC-DOMAIN-EVEN-ROOT · Deepen
D	Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31) SK-FUNC-DOMAIN-FORMULA · Deepen
A	Determine excluded values from an original denominator. (first introduced in Math 21) SK-RAT-RESTRICTIONS · Apply

M31-FUN-005

I can graph a piecewise function from its formula.
TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Describe increasing or decreasing behavior over intervals. (first introduced in Math 31) SK-FUNC-INTERVAL-BEHAVIOR · Apply
I	Select the correct rule of a piecewise function for a given input. SK-FUNC-PIECEWISE-EVALUATE · Introduce
I	Graph each rule on its stated domain with correct endpoints. SK-FUNC-PIECEWISE-GRAPH · Introduce

M31-FUN-006

I can write a piecewise formula that represents a graph.
TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Describe increasing or decreasing behavior over intervals. (first introduced in Math 31) SK-FUNC-INTERVAL-BEHAVIOR · Apply
I	Translate a segmented graph into a piecewise formula. SK-FUNC-PIECEWISE-WRITE · Introduce
A	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Apply

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SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

M31-FUN-007	I can graph a transformed function using shifts, stretches, compressions, and reflections. TEXTBOOK SECTIONS 3.5 Transformation of Functions
SUPPORTING SKILLS	
I	Interpret transformations applied to function inputs. SK-FUNC-TRANSFORM-HORIZONTAL · Introduce
I	Distinguish stretches from compressions and horizontal effects from vertical effects. SK-FUNC-TRANSFORM-SCALE · Introduce
I	Interpret transformations applied to function outputs. SK-FUNC-TRANSFORM-VERTICAL · Introduce
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

M31-FUN-008	I can describe how a transformation affects domain, range, or asymptotes. TEXTBOOK SECTIONS 3.5 Transformation of Functions
SUPPORTING SKILLS	
I	Exchange domain and range between a function and its inverse. SK-FUNC-INVERSE-DOMAIN-RANGE · Introduce
D	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Deepen
D	Distinguish stretches from compressions and horizontal effects from vertical effects. (first introduced in Math 31) SK-FUNC-TRANSFORM-SCALE · Deepen
D	Interpret transformations applied to function outputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-VERTICAL · Deepen

M31-FUN-009	I can evaluate a composition from formulas, graphs, or tables. TEXTBOOK SECTIONS 3.4 Composition of Functions
SUPPORTING SKILLS	
I	Substitute one function formula into another. SK-FUNC-COMPOSE-FORMULA · Introduce
I	Follow intermediate values to compose functions from tables. SK-FUNC-COMPOSE-TABLE · Introduce
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply

M31-FUN-010	I can write a formula for a composition of two functions. TEXTBOOK SECTIONS 3.4 Composition of Functions
SUPPORTING SKILLS	
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
D	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Deepen

M31-FUN-011	I can find an inverse function algebraically. TEXTBOOK SECTIONS 3.7 Inverse Functions
SUPPORTING SKILLS	
A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
I	Exchange variables and solve to obtain an inverse formula. SK-FUNC-INVERSE-ALGEBRA · Introduce
I	Exchange input and output values when interpreting an inverse. SK-FUNC-INVERSE-SWAP · Introduce

M31-FUN-012	I can evaluate and interpret an inverse from a graph, formula, or table. TEXTBOOK SECTIONS 3.7 Inverse Functions
SUPPORTING SKILLS	
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
A	Exchange variables and solve to obtain an inverse formula. (first introduced in Math 31) SK-FUNC-INVERSE-ALGEBRA · Apply
D	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Deepen

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

M31-FUN-013

I can determine and interpret the units of an inverse function.
TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

I

Reverse input and output units when interpreting an inverse.
SK-FUNC-INVERSE-UNITS · Introduce

A

Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12)
SK-REP-UNITS · Apply

M31-FUN-014

I can explain how the domain and range of a function relate to those of its inverse.
TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

I

Exchange domain and range between a function and its inverse.
SK-FUNC-INVERSE-DOMAIN-RANGE · Introduce

A

Exchange input and output values when interpreting an inverse. (first introduced in Math 31)
SK-FUNC-INVERSE-SWAP · Apply

M31-FUN-015

I can determine the range of a function by analyzing the domain of its inverse.
TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

A

Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31)
SK-FUNC-DOMAIN-FORMULA · Apply

D

Exchange domain and range between a function and its inverse. (first introduced in Math 31)
SK-FUNC-INVERSE-DOMAIN-RANGE · Deepen

M31-FUN-016

I can verify algebraically that two functions are inverses.
TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

A

Simplify an algebraic expression using operation properties. (inherited from middle school)
SK-ALG-EXPRESSION-SIMPLIFY · Apply

A

Substitute one function formula into another. (first introduced in Math 31)
SK-FUNC-COMPOSE-FORMULA · Apply

I

Verify inverse functions using composition.
SK-FUNC-INVERSE-VERIFY · Introduce

M31-FUN-017

I can interpret the meaning and units of a composition.
TEXTBOOK SECTIONS 3.4 Composition of Functions

SUPPORTING SKILLS

D

Substitute one function formula into another. (first introduced in Math 31)
SK-FUNC-COMPOSE-FORMULA · Deepen

D

Follow intermediate values to compose functions from tables. (first introduced in Math 31)
SK-FUNC-COMPOSE-TABLE · Deepen

A

Explain a model parameter in the language of its context. (first introduced in Math 21)
SK-MODEL-PARAMETER-INTERPRET · Apply

D

Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12)
SK-REP-UNITS · Deepen

M31-FUN-018

I can classify a graph or table as concave up or concave down using changes in average rate of change.
TEXTBOOK SECTIONS 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

A

Calculate average rate of change over an interval. (first introduced in Math 21)
SK-FUNC-AROC · Apply

I

Use successive average rates of change to classify concavity.
SK-FUNC-CONCAVITY-AROC · Introduce

D

Use tabular patterns to predict graph shape and behavior. (first introduced in Math 21)
SK-REP-TABLE-GRAPH · Deepen

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M31-FUN-019

I can determine whether a function is invertible using the horizontal line test.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

I

Apply the horizontal line test for invertibility.
SK-FUNC-INVERSE-HLT · Introduce

I

Test whether a function is one-to-one on a domain.
SK-FUNC-ONE-TO-ONE · Introduce

M31-FUN-020

I can graph an inverse function as a reflection across the line $y = x$.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

A

Exchange domain and range between a function and its inverse. (first introduced in Math 31)
SK-FUNC-INVERSE-DOMAIN-RANGE · Apply

D

Exchange input and output values when interpreting an inverse. (first introduced in Math 31)
SK-FUNC-INVERSE-SWAP · Deepen

D

Connect parameters in a formula to features of its graph. (first introduced in Math 21)
SK-REP-FORMULA-GRAPH · Deepen

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SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Quadratic Functions and Equations

REQUIRED

UNIT TEXTBOOK COVERAGE

2.5 Quadratic Equations; 5.1 Quadratic Functions

M31-QUA D-001	I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function. TEXTBOOK SECTIONS 5.1 Quadratic Functions
SUPPORTING SKILLS	
A	Use successive average rates of change to classify concavity. (first introduced in Math 31) SK-FUNC-CONCAVITY-AROC · Apply
A	Identify standard, factored, and vertex forms. (first introduced in Math 21) SK-QUAD-FORM-IDENTIFY · Apply
D	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Deepen
D	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Deepen

M31-QUA D-002	I can connect standard, factored, and vertex forms to features of a quadratic graph. TEXTBOOK SECTIONS 5.1 Quadratic Functions
SUPPORTING SKILLS	
D	Identify standard, factored, and vertex forms. (first introduced in Math 21) SK-QUAD-FORM-IDENTIFY · Deepen
A	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Apply
A	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Apply
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-QUA D-003	I can rewrite a quadratic function in a form that reveals a requested feature. TEXTBOOK SECTIONS 5.1 Quadratic Functions
SUPPORTING SKILLS	
A	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Apply
A	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Apply
D	Create a perfect-square trinomial by completing the square. (first introduced in Math 21) SK-QUAD-COMPLETE-SQUARE · Deepen
D	Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method. (first introduced in Math 21) SK-QUAD-FORM-CONVERT · Deepen

M31-QUA D-004	I can solve a quadratic equation by factoring or using the quadratic formula. TEXTBOOK SECTIONS 2.5 Quadratic Equations; 5.1 Quadratic Functions
SUPPORTING SKILLS	
A	Apply the zero-product property. (first introduced in Math 21) SK-FACTOR-ZERO-PRODUCT · Apply
A	Use the discriminant to predict the number and type of solutions. (first introduced in Math 21) SK-QUAD-DISCRIMINANT · Apply
A	Substitute coefficients accurately into the quadratic formula. (first introduced in Math 21) SK-QUAD-FORMULA · Apply
D	Select an efficient quadratic-solving method from equation structure. (first introduced in Math 21) SK-QUAD-METHOD-SELECT · Deepen

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

M31-QUA D-005	I can graph a quadratic function using its algebraic structure and key features. TEXTBOOK SECTIONS 5.1 Quadratic Functions
SUPPORTING SKILLS	
A	Identify standard, factored, and vertex forms. (first introduced in Math 21) SK-QUAD-FORM-IDENTIFY · Apply
D	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Deepen
D	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Deepen
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-QUA D-006	I can construct a quadratic function from specified zeros, a vertex, or graphical information. TEXTBOOK SECTIONS 5.1 Quadratic Functions
SUPPORTING SKILLS	
A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply
I	Build a polynomial from zeros, multiplicities, and a point. SK-POLY-CONSTRUCT · Introduce
A	Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method. (first introduced in Math 21) SK-QUAD-FORM-CONVERT · Apply

M31-QUA D-007	I can create and interpret a quadratic trajectory model. TEXTBOOK SECTIONS 5.1 Quadratic Functions
SUPPORTING SKILLS	
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
D	Interpret vertex and zeros in an application. (first introduced in Math 21) SK-QUAD-MODEL-FEATURES · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Exponential Functions and Models

REQUIRED

UNIT TEXTBOOK COVERAGE

6.1 Exponential Functions; 6.2 Graphs of Exponential Functions; 6.7 Exponential and Logarithmic Models

M31-EXP-001

I can distinguish linear change from exponential change using differences, ratios, or percent change.
TEXTBOOK SECTIONS 6.1 Exponential Functions

SUPPORTING SKILLS

D	Use constant differences or ratios to distinguish linear and exponential patterns. (first introduced in Math 31) SK-EXP-DIFFERENCE-RATIO · Deepen
D	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Deepen
A	Calculate and interpret percent increase or decrease. (inherited from middle school) SK-NUM-PERCENT-CHANGE · Apply

M31-EXP-002

I can identify initial value and growth or decay factor in an exponential model.
TEXTBOOK SECTIONS 6.1 Exponential Functions; 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

I	Identify initial amount, rate, and time interval in context. SK-EXP-MODEL-CONTEXT · Introduce
I	Convert a percent increase or decrease to a growth or decay factor. SK-EXP-PERCENT-FACTOR · Introduce
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen

M31-EXP-003

I can write an exponential function from a table, graph, or context.
TEXTBOOK SECTIONS 6.1 Exponential Functions

SUPPORTING SKILLS

D	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Deepen
I	Determine an exponential model from tabular values. SK-EXP-MODEL-TABLE · Introduce
I	Solve for an exponential model using two points. SK-EXP-MODEL-TWO-POINTS · Introduce
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-EXP-004

I can evaluate and interpret an exponential model.
TEXTBOOK SECTIONS 6.1 Exponential Functions; 6.2 Graphs of Exponential Functions; 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

D	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Deepen
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M31-EXP-005

I can calculate growth under periodic compounding.
TEXTBOOK SECTIONS 6.1 Exponential Functions; 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

I	Substitute correctly into a periodic-compounding model. SK-EXP-COMPOUND-PERIODIC · Introduce
A	Convert a percent increase or decrease to a growth or decay factor. (first introduced in Math 31) SK-EXP-PERCENT-FACTOR · Apply
A	Calculate a percentage of a quantity or determine what percent one quantity is of another. (inherited from middle school) SK-NUM-PERCENT-CALCULATE · Apply

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

M31-EXP-006

I can calculate growth under continuous compounding.
TEXTBOOK SECTIONS 6.1 Exponential Functions; 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

I

Substitute correctly into a continuous-compounding model.
SK-EXP-COMPOUND-CONTINUOUS · Introduce

A

Identify initial amount, rate, and time interval in context. (first introduced in Math 31)
SK-EXP-MODEL-CONTEXT · Apply

M31-EXP-007

I can determine an exponential formula from two points or other sufficient data.
TEXTBOOK SECTIONS 6.1 Exponential Functions

SUPPORTING SKILLS

A

Isolate a specified variable in an equation or formula. (first introduced in Math 12)
SK-ALG-VARIABLE-ISOLATE · Apply

D

Solve for an exponential model using two points. (first introduced in Math 31)
SK-EXP-MODEL-TWO-POINTS · Deepen

A

Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21)
SK-EXP-RATIONAL · Apply

M31-EXP-008

I can determine domain, range, intercepts, and asymptotes of an exponential function.
TEXTBOOK SECTIONS 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

I

Identify horizontal asymptotic behavior of an exponential graph.
SK-EXP-ASYMPTOTE · Introduce

D

Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31)
SK-FUNC-DOMAIN-FORMULA · Deepen

D

Connect parameters in a formula to features of its graph. (first introduced in Math 21)
SK-REP-FORMULA-GRAPH · Deepen

M31-EXP-009

I can compare the growth rates of two exponential functions.
TEXTBOOK SECTIONS 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

I

Compare exponential growth rates expressed in different forms.
SK-EXP-COMPARE-GROWTH · Introduce

D

Compare functions shown in different representations. (first introduced in Math 21)
SK-FUNC-COMPARE-REP · Deepen

M31-EXP-010

I can determine the percent growth or decay represented by a continuous exponential model.
TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

I

Convert between equivalent discrete- and continuous-growth forms.
SK-EXP-DISCRETE-CONTINUOUS · Introduce

D

Convert a percent increase or decrease to a growth or decay factor. (first introduced in Math 31)
SK-EXP-PERCENT-FACTOR · Deepen

A

Calculate and interpret percent increase or decrease. (inherited from middle school)
SK-NUM-PERCENT-CHANGE · Apply

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Logarithmic Functions and Models

REQUIRED

UNIT TEXTBOOK COVERAGE

6.3 Logarithmic Functions; 6.4 Graphs of Logarithmic Functions; 6.5 Logarithmic Properties; 6.6 Exponential and Logarithmic Equations; 6.7 Exponential and Logarithmic Models

M31-LOG-001

I can convert between logarithmic and exponential forms.
TEXTBOOK SECTIONS 6.3 Logarithmic Functions

SUPPORTING SKILLS

I	Convert between logarithmic and exponential forms. SK-LOG-CONVERT · Introduce
I	Interpret a logarithm as an exponent. SK-LOG-DEFINITION · Introduce

M31-LOG-002

I can evaluate a logarithm using its exponential meaning.
TEXTBOOK SECTIONS 6.3 Logarithmic Functions

SUPPORTING SKILLS

I	Rewrite exponential expressions using a common base when possible. SK-EXP-REWRITE-BASE · Introduce
D	Interpret a logarithm as an exponent. (first introduced in Math 31) SK-LOG-DEFINITION · Deepen

M31-LOG-003

I can expand or condense an expression using product, quotient, and power properties.
TEXTBOOK SECTIONS 6.5 Logarithmic Properties

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Apply the logarithm power property in either direction. SK-LOG-POWER · Introduce
I	Apply the logarithm product property in either direction. SK-LOG-PRODUCT · Introduce
I	Apply the logarithm quotient property in either direction. SK-LOG-QUOTIENT · Introduce

M31-LOG-004

I can solve an exponential equation using logarithms.
TEXTBOOK SECTIONS 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
A	Apply the logarithm power property in either direction. (first introduced in Math 31) SK-LOG-POWER · Apply
I	Isolate an exponential expression and apply a logarithm. SK-LOG-SOLVE-EXP · Introduce

M31-LOG-005

I can solve a logarithmic equation using exponential form and check domain restrictions.
TEXTBOOK SECTIONS 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

A	Convert between logarithmic and exponential forms. (first introduced in Math 31) SK-LOG-CONVERT · Apply
I	Reject solutions that make a logarithm argument nonpositive. SK-LOG-DOMAIN-CHECK · Introduce
I	Combine or isolate logarithms before converting to exponential form. SK-LOG-SOLVE-LOG · Introduce

M31-LOG-006

I can calculate and interpret doubling time or half-life.
TEXTBOOK SECTIONS 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

I	Translate a doubling-time or half-life question into an exponential equation. SK-LOG-DOUBLING-HALF · Introduce
A	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M31-LOG-007

I can determine domain, range, intercepts, and asymptotes of a logarithmic function.
TEXTBOOK SECTIONS 6.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

D	Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31) SK-FUNC-DOMAIN-FORMULA · Deepen
I	Identify vertical asymptotic behavior of a logarithmic graph. SK-LOG-ASYMPTOTE · Introduce
A	Reject solutions that make a logarithm argument nonpositive. (first introduced in Math 31) SK-LOG-DOMAIN-CHECK · Apply
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-LOG-008

I can explain how exponential and logarithmic functions undo one another.
TEXTBOOK SECTIONS 6.3 Logarithmic Functions; 6.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

A	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Apply
A	Verify inverse functions using composition. (first introduced in Math 31) SK-FUNC-INVERSE-VERIFY · Apply
D	Interpret a logarithm as an exponent. (first introduced in Math 31) SK-LOG-DEFINITION · Deepen

M31-LOG-009

I can solve a multi-step exponential or logarithmic equation.
TEXTBOOK SECTIONS 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

A	Reject solutions that make a logarithm argument nonpositive. (first introduced in Math 31) SK-LOG-DOMAIN-CHECK · Apply
A	Apply the logarithm power property in either direction. (first introduced in Math 31) SK-LOG-POWER · Apply
A	Apply the logarithm product property in either direction. (first introduced in Math 31) SK-LOG-PRODUCT · Apply
A	Apply the logarithm quotient property in either direction. (first introduced in Math 31) SK-LOG-QUOTIENT · Apply
D	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Deepen
D	Combine or isolate logarithms before converting to exponential form. (first introduced in Math 31) SK-LOG-SOLVE-LOG · Deepen

M31-LOG-010

I can determine an intersection of exponential functions graphically or algebraically.
TEXTBOOK SECTIONS 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

A	Compare exponential growth rates expressed in different forms. (first introduced in Math 31) SK-EXP-COMPARE-GROWTH · Apply
D	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Deepen
A	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Apply
A	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Apply

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M31-LOG-011	I can rewrite a discrete exponential model as an equivalent continuous exponential model. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
D	Convert between equivalent discrete- and continuous-growth forms. (first introduced in Math 31) SK-EXP-DISCRETE-CONTINUOUS · Deepen
A	Convert a percent increase or decrease to a growth or decay factor. (first introduced in Math 31) SK-EXP-PERCENT-FACTOR · Apply
A	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Apply

M31-LOG-012	I can solve and interpret an exponential or logarithmic modeling problem. TEXTBOOK SECTIONS 6.7 Exponential and Logarithmic Models
SUPPORTING SKILLS	
D	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Deepen
A	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Apply
A	Combine or isolate logarithms before converting to exponential form. (first introduced in Math 31) SK-LOG-SOLVE-LOG · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Extension Content

EXTENSION

UNIT TEXTBOOK COVERAGE

3.6 Absolute Value Functions

M31-EXT-001	I can analyze an absolute-value function using transformations. TEXTBOOK SECTIONS 3.6 Absolute Value Functions
SUPPORTING SKILLS	
A	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Apply
A	Distinguish stretches from compressions and horizontal effects from vertical effects. (first introduced in Math 31) SK-FUNC-TRANSFORM-SCALE · Apply
A	Interpret transformations applied to function outputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-VERTICAL · Apply
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

M31-EXT-002	I can determine even or odd symmetry from a graph or formula. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Test a formula or graph for even or odd symmetry. SK-FUNC-SYMMETRY · Introduce

M31-EXT-003	I can interpret a contextual logarithmic scale. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
A	Interpret a logarithm as an exponent. (first introduced in Math 31) SK-LOG-DEFINITION · Apply
I	Interpret equal intervals on a logarithmic scale as equal multiplicative changes. SK-LOG-SCALE-INTERPRET · Introduce
A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply